

Problem 4

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Problem 1 Compute the following integrals:

(a) $\int_0^1 e^{5x} dx$

Explanation. $\int_0^1 e^{5x} dx = \left[\frac{1}{5} e^{5x} \right]_0^1 = \frac{1}{5} (e^5 - e^0) = \frac{1}{5} (e^5 - 1)$

(b) $\int_{-2}^{-1} \frac{1}{x^3} dx$

Explanation.

$$\begin{aligned} \int_{-2}^{-1} \frac{1}{x^3} dx &= \int_{-2}^{-1} x^{-3} dx \\ &= \left[\frac{x^{-2}}{-2} \right]_{-2}^{-1} \\ &= \left[\frac{-1}{2x^2} \right]_{-2}^{-1} \\ &= -\frac{1}{2} - \left(-\frac{1}{8} \right) \\ &= -\frac{1}{2} + \frac{1}{8} = -\frac{3}{8} \end{aligned}$$

(c) $\int_0^4 (3x - 5 + 7\sqrt{16 - x^2}) dx$

Explanation. First notice that:

$$\int_0^4 (3x - 5 + 7\sqrt{16 - x^2}) dx = \int_0^4 (3x - 5) dx + 7 \int_0^4 \sqrt{16 - x^2} dx. \quad (1)$$

To get the easy part out of the way first:

$$\int_0^4 (3x - 5) dx = \left[\frac{3}{2} x^2 - 5x \right]_0^4 = (24 - 20) - (0 - 0) = 4. \quad (2)$$

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So the real issue is in computing $\int_0^4 \sqrt{16 - x^2} dx$.

We do not know how to integrate $\sqrt{16 - x^2}$ directly, but recall that the solution set to the equation $x^2 + y^2 = 16$ is a circle with radius 4 centered at the origin. Solving for y we get $y = \pm\sqrt{16 - x^2}$. So if we restrict ourselves to $y = \sqrt{16 - x^2}$ and $0 \leq x \leq 4$, this gives us the upper-right quarter of the circle (draw a picture and convince yourself of this). Since the total area of the circle of $\pi(4)^2 = 16\pi$, the area under this portion of the curve is $\frac{1}{4}(16\pi) = 4\pi$. Thus we have computed (geometrically) that

$$\int_0^4 \sqrt{16 - x^2} dx = 4\pi. \quad (3)$$

So, $\int_0^4 (3x - 5 + 7\sqrt{16 - x^2}) dx = 4 + 7(4\pi) = 4 + 28\pi$ by (1),(2), and (3)

[3]